

DENTINOGENESIS

DR. UME HANI

LECTURER- DEPT OF ORAL BIOLOGY



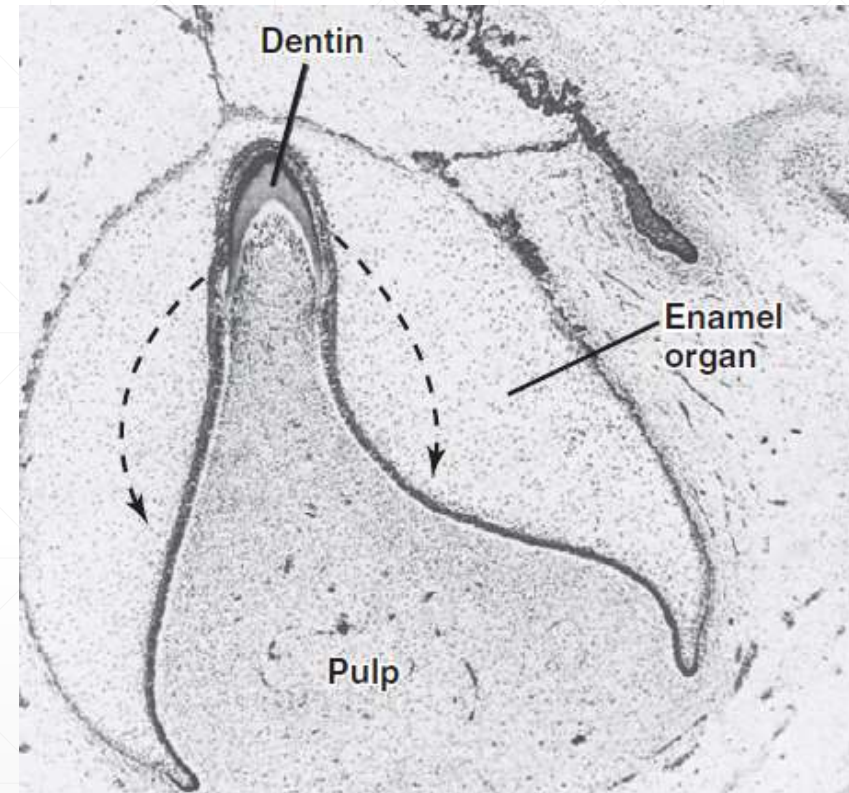
LEARNING OBJECTIVES

By the end of this topic, the students will be able to;

- Define dentinogenesis and explain its stages.
 - Describe the differentiation of odontoblasts and their role in dentine formation.
 - Explain the role of dentine phosphoprotein in mineralization.
 - Compare the types and patterns of dentine mineralization.
 - Differentiate between mantle dentine and circumpulpal dentine with respect to structure and mineralization.
 - Explain the formation and significance of calcospherites.
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DENTINOGENESIS

- Dentinogenesis is the process by which odontoblasts form dentine. It begins at the late bell stage of tooth development after differentiation of odontoblasts from the dental papilla around 18 weeks of development.
- Dentine formation, as for bone and cementum, is a two-phase process, beginning with the secretion of an extracellular connective tissue matrix and its subsequent mineralization.



Dentin formation during the bell stage of tooth development.

STAGES OF DENTINOGENESIS

Differentiation of odontoblasts



Deposition of organic matrix
(Predentine formation)



Mineralization of organic matrix
(Primary dentine formation)



Secondary and tertiary dentine
formation

LIFE CYCLE OF ODONTOBLAST

- Basal lamina
- Endoplasmic reticulum
- Mitochondria
- Golgi apparatus
- Secretory vesicles
- Dentine
- Initial enamel matrix
- Enamel crystallites

Cells of IEE start to differentiate first



Ectomesenchymal cells of dental papilla

Daughter cells

Mitosis

Fully differentiated odontoblast

First layer of dentine

Subodontoblast

1

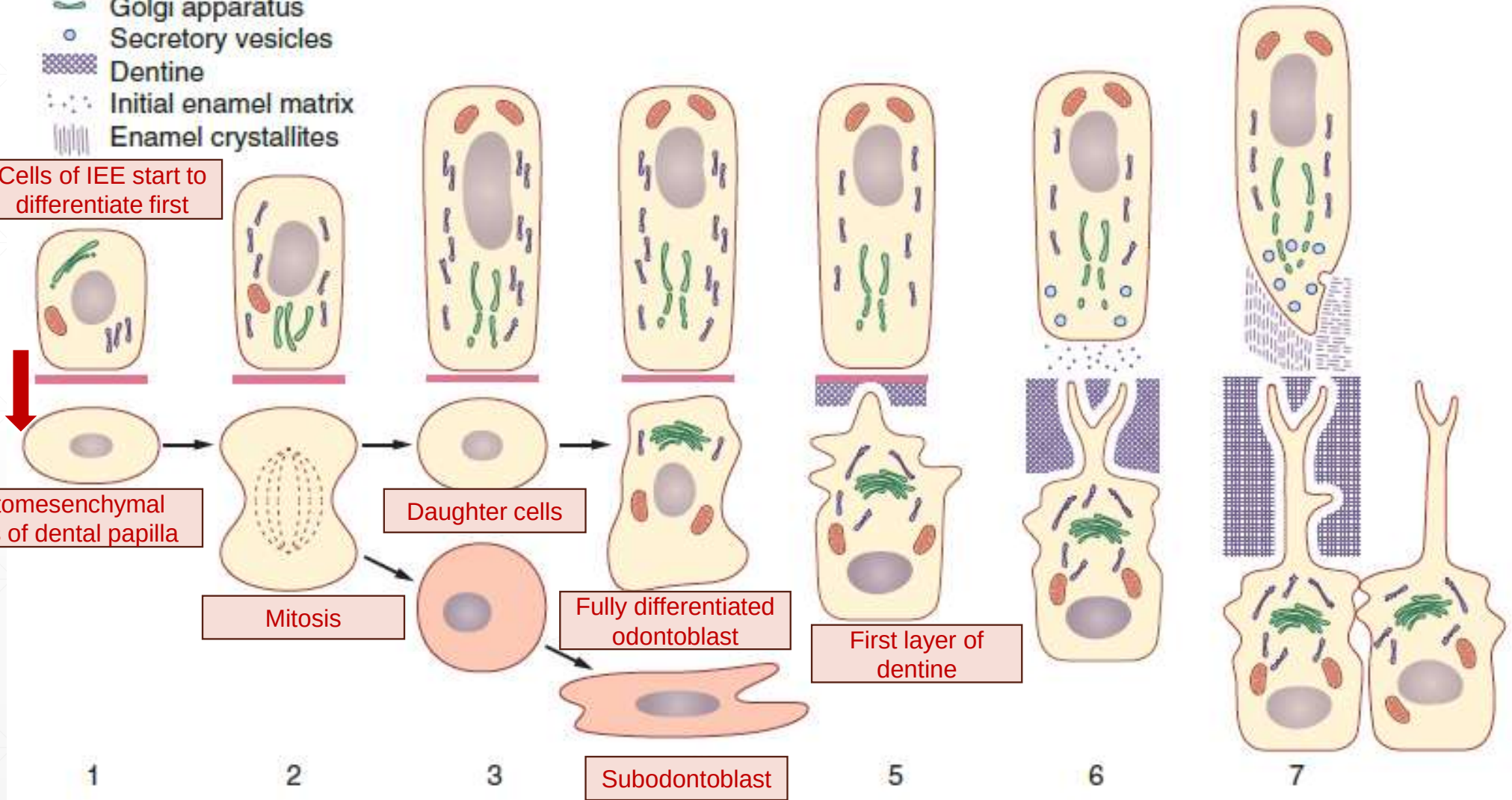
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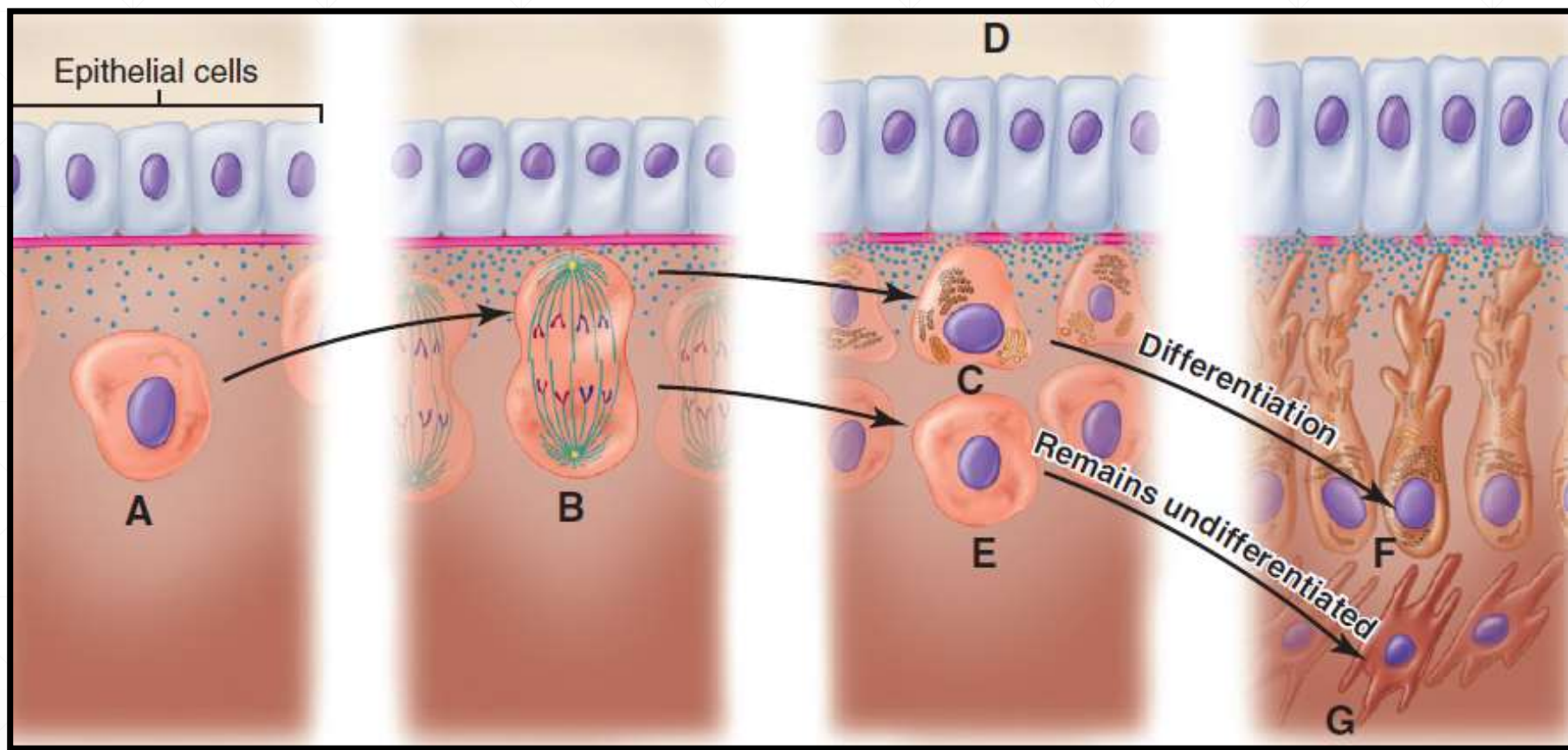
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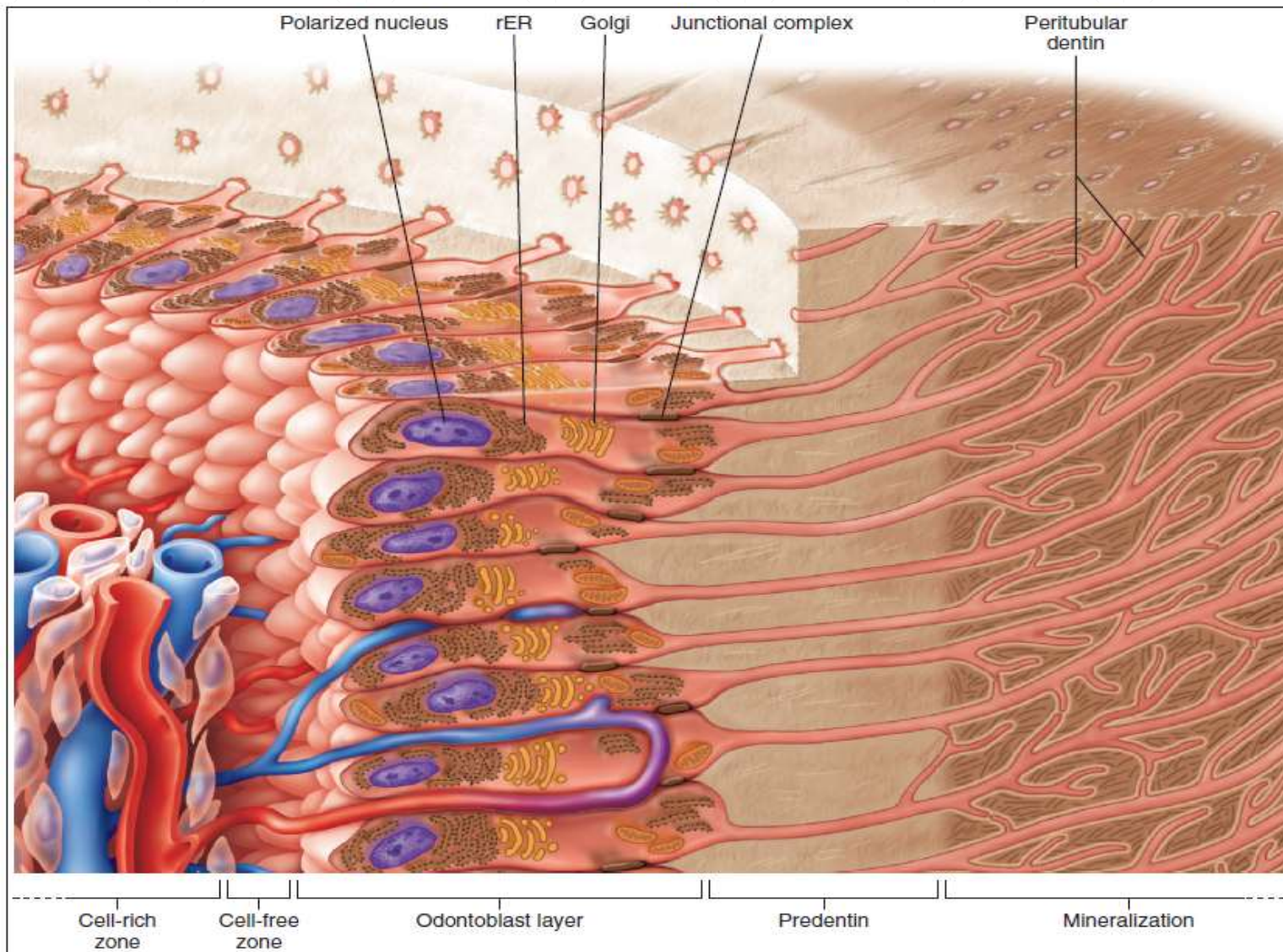


DIFFERENTIATION OF ODONTOBLASTS

- **Neural crest–derived ectomesenchymal cells of the dental papilla become odontoblasts:** During the late bell stage, dental papilla cells differentiate into odontoblasts under the influence of signals from the internal enamel epithelium.
 - **Reciprocal epithelial–mesenchymal signaling controls differentiation:** BMP, FGF, Shh, Wnt, TGF- β , and IGF. Continuous two-way communication between enamel epithelium and dental papilla regulates tooth development. Enamel knot acts as signaling center and induces odontoblast differentiation.
-



A: an undifferentiated ectomesenchymal cell of the dental papilla divides.
 B: mitotic activity, C: daughter cell influenced by the epithelial cells. F: differentiated odontoblasts,
 E: another daughter cell not exposed to this epithelial influence, G: subodontoblast cell



1. Released from basement membrane.
Stimulate preodontoblast differentiation.

2. Regulate cytoskeletal organization
and cell shape changes.

**TGF- β ROLE IN
DIFFERENTIATION OF
ODONTOBLASTS**

3. Promote relocation of intracellular
organelles.

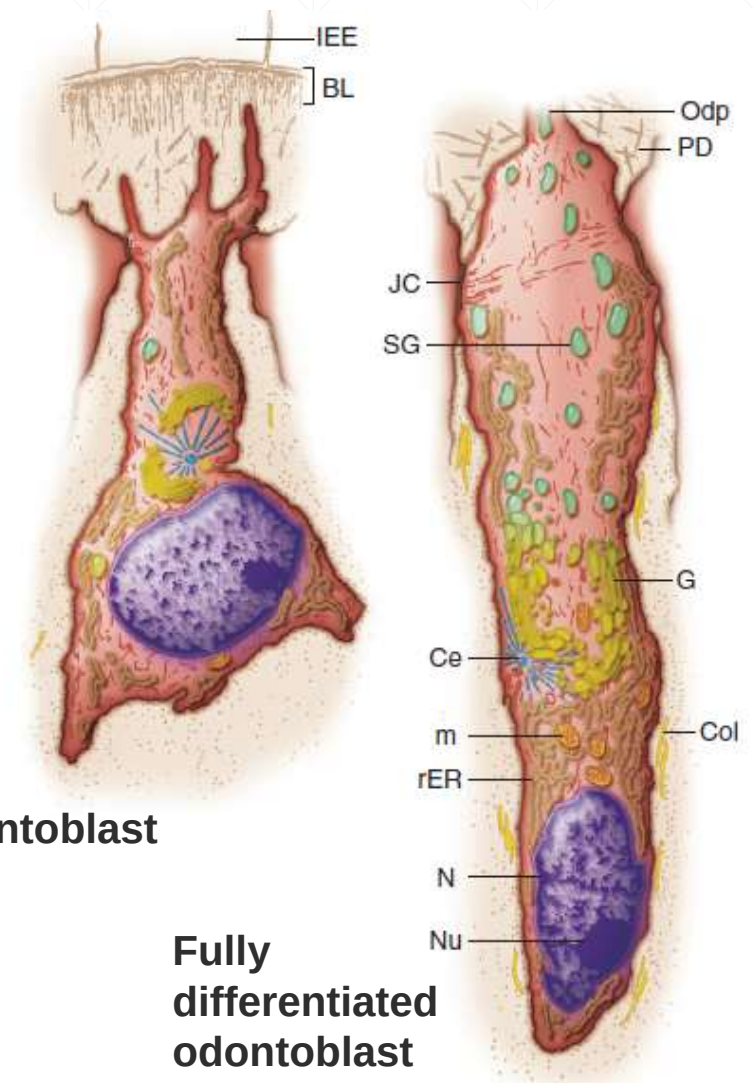
4. Help cells exit the mitotic cycle and
become mature odontoblasts.

MORPHOLOGICAL CHANGES OF ODONTOBLASTS

- This begins with the dental papilla cells adjacent to the deepest invaginations of the internal enamel epithelium, the cusps or incisal margins.
- The preodontoblasts then increase in size (hypertrophy) and the nucleus comes to lie in the basal part of the cell, furthest from the IEE, the Golgi complex becomes pronounced and positioned above the nucleus. RER also increases in size.
- Cell elongation and polarization occur by redistribution of actin, vinculin, and vimentin.

Preodontoblast

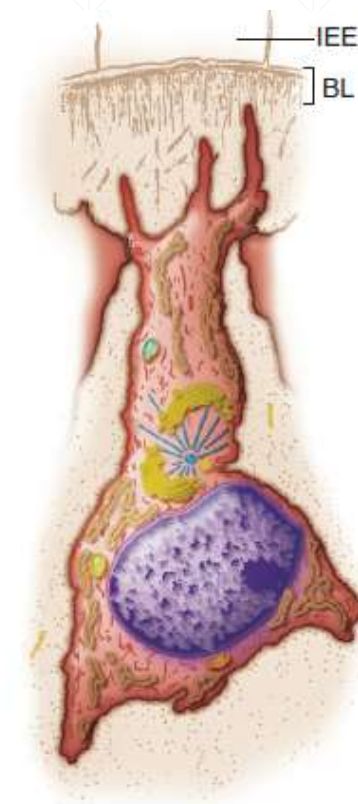
Fully differentiated odontoblast



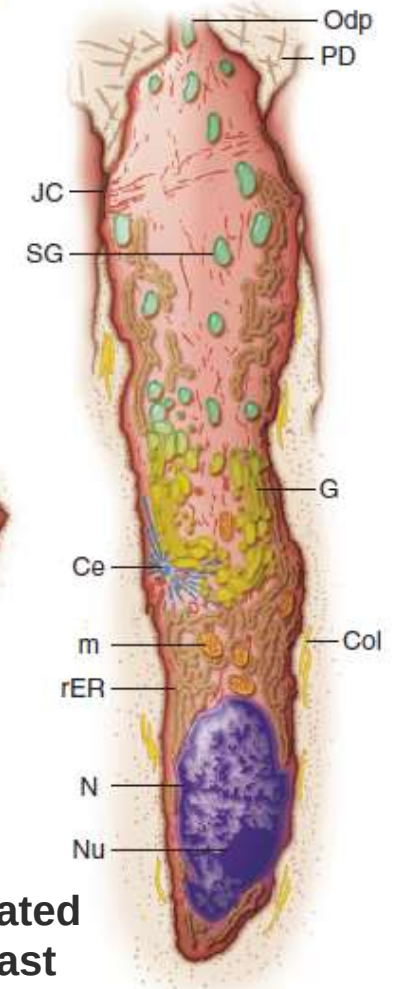
MORPHOLOGICAL CHANGES OF ODONTOBLASTS

- Many processes extend from odontoblasts, but most processes regress and one dominant odontoblastic process remains.
- Odontoblast differentiation progresses from cusp tip/incisal margin towards the root and the mature odontoblasts begin dentinogenesis.
- During dentine mineralization the transcellular calcium transport through odontoblasts (via L-type voltage-gated calcium channels).

Preodontoblast

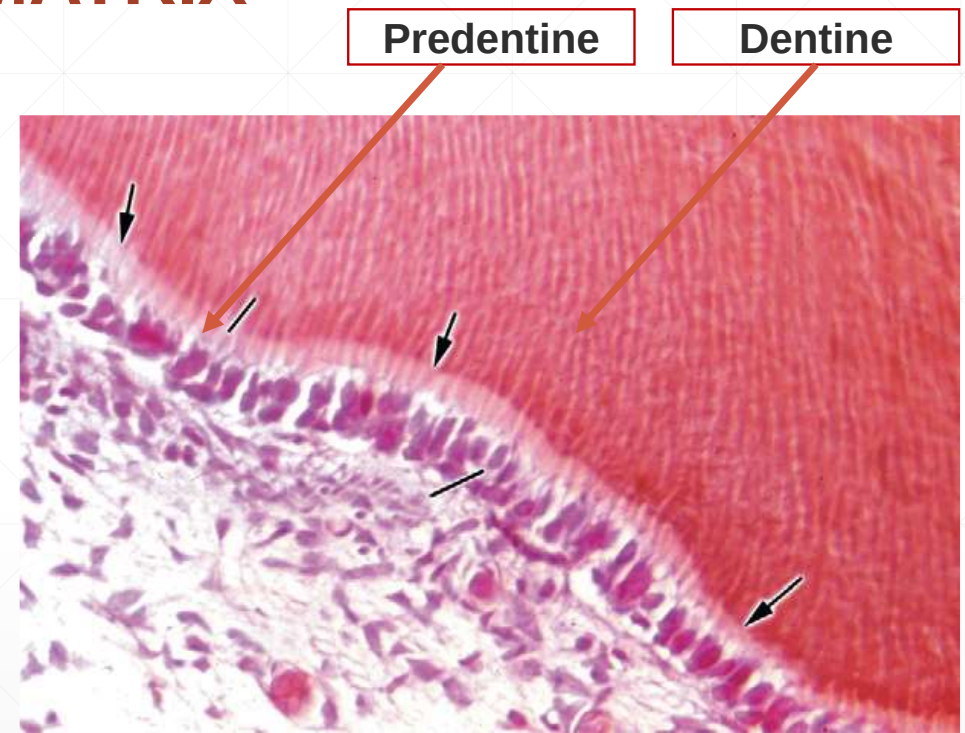


Fully differentiated odontoblast



DEPOSITION OF DENTINE MATRIX

- The mature functioning odontoblasts secrete the organic matrix of dentine.
- These cells synthesize Type I Collagen and DPP and precursors are found within the RER, Golgi complex and secretory granules (different intracellular compartments).
- The matrix thus formed consists primarily of type I collagen fibrils with DPP as the second most abundant constituent.
- The odontoblast process secretes organic matrix such that a complex of collagen and non-collagenous proteins is formed at the junction of the predentine and calcified dentine.



Light photomicrograph of the predentin-dentin interface showing mineralization

Table 23.1 Components of organic dentine extracellular matrix**Collagens:**

Type 1

Type 1 trimer

Type V

Proteoglycans:

Decorin (PG II)

Biglycan (PG I)

Other chondroitin 4-sulphate-containing proteoglycans

Keratan sulphate (?)

Glycoproteins/sialoproteins:

Osteonectin

Dentine sialoprotein(s) (DSPs)

Bone sialoprotein

Osteopontin

Bone acidic glycoprotein 75

Syndecan 2

Phosphoproteins:

Dentine phosphoproteins (DPPs)

Dentine matrix protein 1

 γ -Carboxyglutamate-containing proteins

Osteocalcin

Amelin-1 (transient expression)

Enamelysin:**Lipids:**

Cholesterol

Cholesterol esters

Triacylglycerols

Serum-derived proteins: α 2HS-glycoprotein

Albumin

Immunoglobulins

Growth factors:TGF β s

Chondrogenic inducing factor

Bone morphogenic proteins (BMPs)

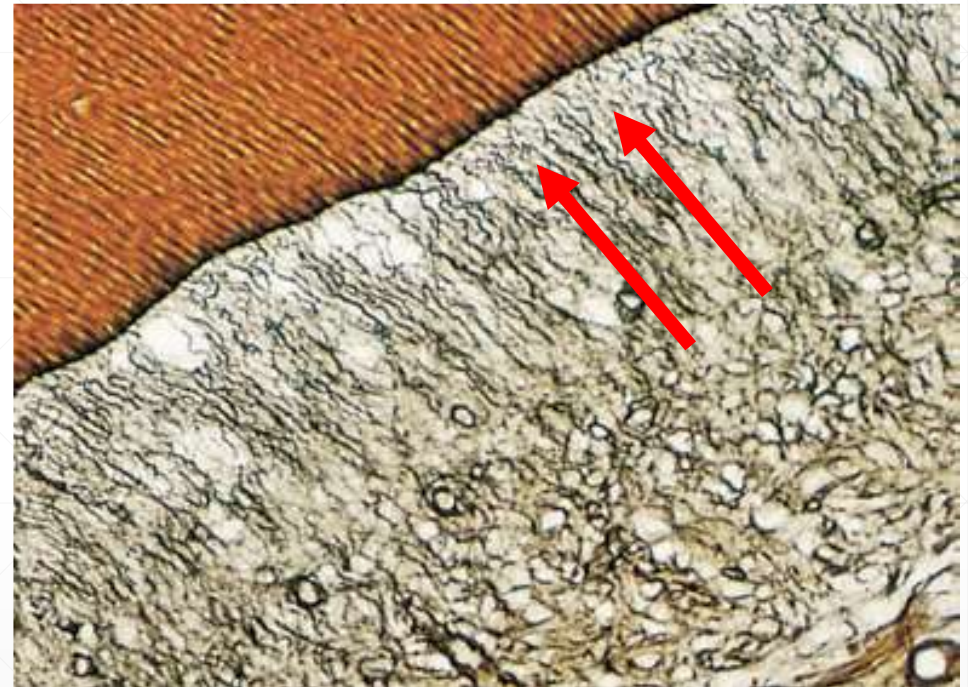
Fibroblast growth factors (FGFs)

Insulin-like growth factors (IGFs)

Sonic hedgehog (SHH)

Collagen Type I

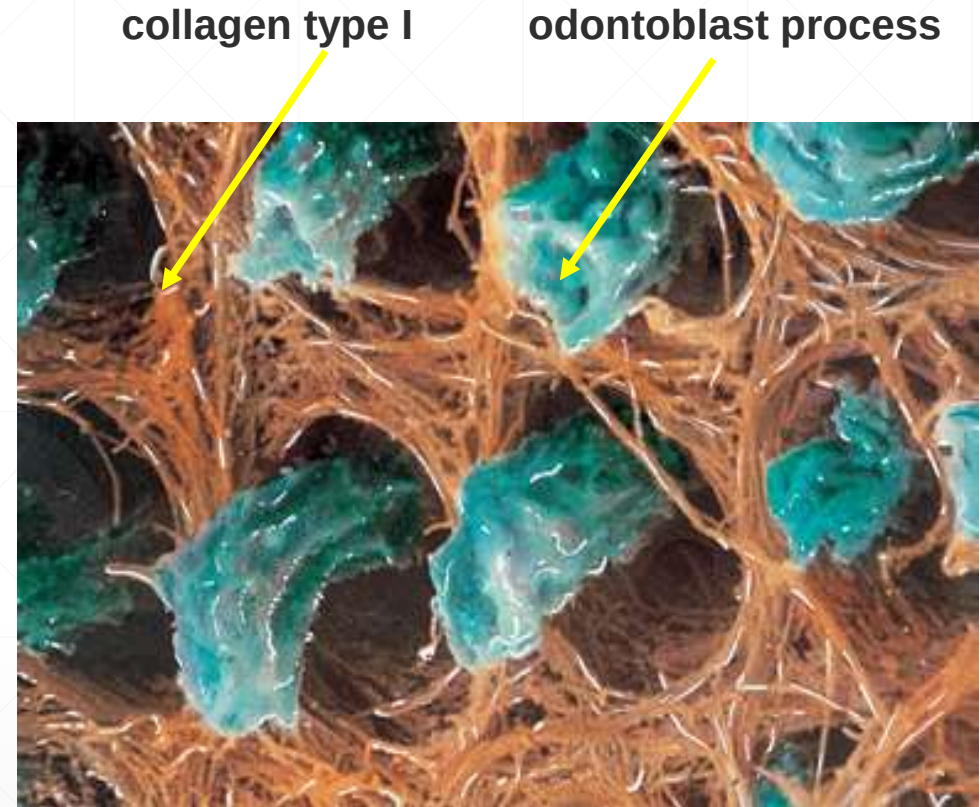
- Collagen I is localised within rosette-form endoplasmic reticulum compartments, the Golgi complex and large, distinctive secretory granules.
- Collagen type I is deposited at the cell border, the cell–predentine boundary by the odontoblast cell body.
- The fibrils laid down initially (mantle dentine) lie at right angles (perpendicular) to the future DEJ, these fibres take on a **‘corkscrew’ appearance** and are called Von Korff’s fibers.



Micrograph of a demineralised section showing 'corkscrew' fibres (von Korff's) shown by arrow.

Collagen Type I

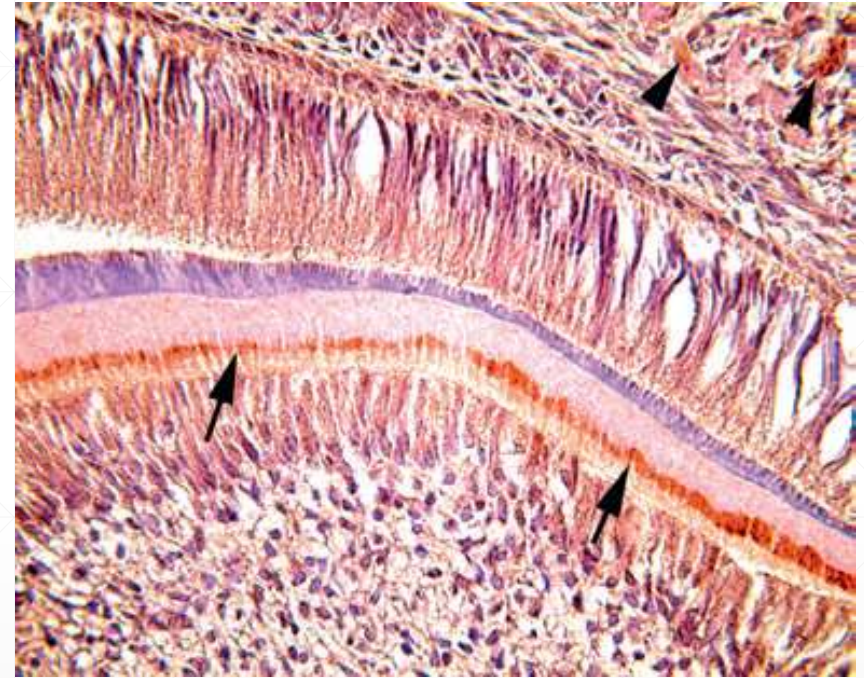
- Once the mantle layer is formed, the remaining type I collagen is laid down with its fibrils approximately parallel to the pulp–dentine border but perpendicular to odontoblast process.



SEM showing the surface of predentine. The collagen fibrils form an interlacing network perpendicular to the odontoblast process.

Dentine Phosphoprotein

- DPP is localised within the tubular endoplasmic reticulum, round-shaped transitional vesicles, the Golgi complex and in narrow asymmetric secretory granules.
- DPP is absent from nonmineralised predentine matrix and concentrated at the mineralisation front. It is released from the odontoblast process.
- DPP binds in a highly specific manner at the 'e' band in the gap region of the type I fibrils, it then attracts calcium ions and helps nucleate and regulate hydroxyapatite crystal growth.



Localisation of phosphoprotein at the mineralisation front (arrows) and in the odontoblast cells

1. Involved in the initiation and growth of mineral and its distribution.

2. Master of mineralization and major role in 'heterogenous nucleation'.

ROLE OF DENTINE PHOSPHOPROTEIN

3. Highly anionic and thus bind increasing numbers of calcium ions, allowing the formation and growth of crystal.

4. Can induce hydroxyapatite formation at physiological ionic concentrations.

Type I collagen is secreted by the odontoblast cell body. As the odontoblast moves pulpally (inward), it continuously deposits collagen-rich matrix forming predentine.

Dentin Phosphoprotein is secreted mainly from the odontoblast process. The odontoblast process extends through the predentine toward the mineralization front. DPP is released some distance away from the cell body, it bypasses much of the predentine and reach the mineralization front.

MINERALIZATION OF DENTINE

- The key element in initiating and controlling mineralization is clearly the odontoblast, by secreting the matrix that becomes mineralized, the transport and release of calcium ions. It determines the presence and distribution of the matrix components that can initiate and modulate the process.
- Mineralization only occurs when odontoblasts are present.



TYPES OF MINERALIZATION

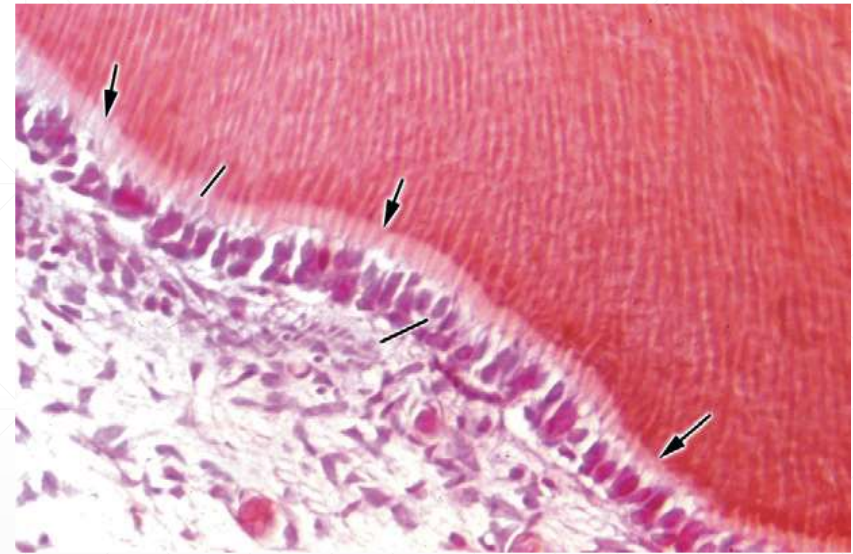
- Heterogenous nucleation
- Matrix vesicle mediated

PATTERNS OF MINERALIZATION

- Linear pattern
 - Globular pattern
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MINERALIZATION BY HETEROGENEOUS NUCLEATION

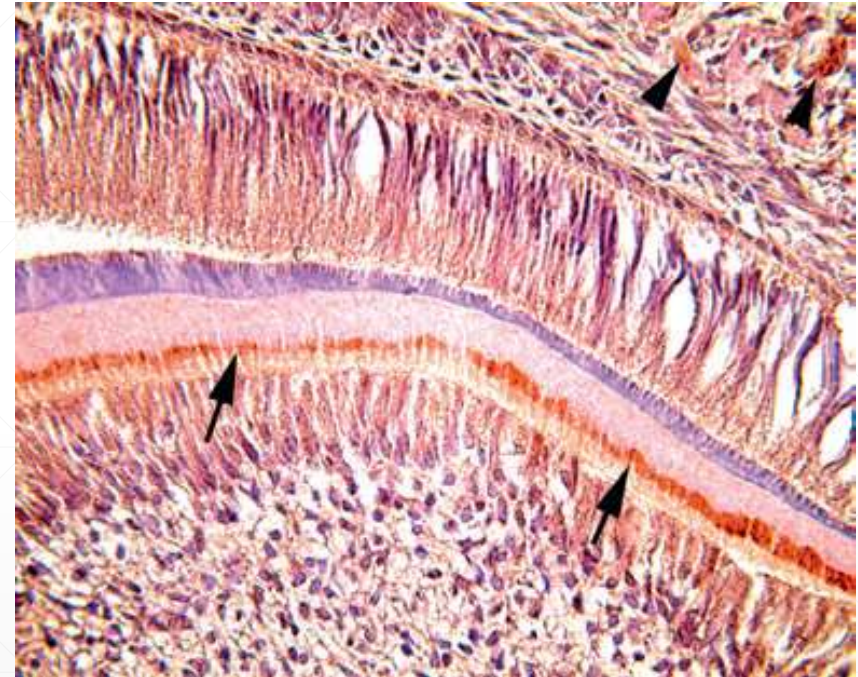
- Dentine mineralization occurs primarily by **heterogeneous nucleation**, in which hydroxyapatite crystals are deposited on the collagen matrix (mediated by DPP).
- The calcium transported by the odontoblasts becomes a crystalline mineral in the dentine by deposition onto a template formed by type I collagen fibrils. It is largely under the control of DPP.



Demineralised section showing the region of mineralising dentine. A = predentine; B = mineralized dentine; C = calcospherites

MINERALIZATION BY HETEROGENOUS NUCLEATION

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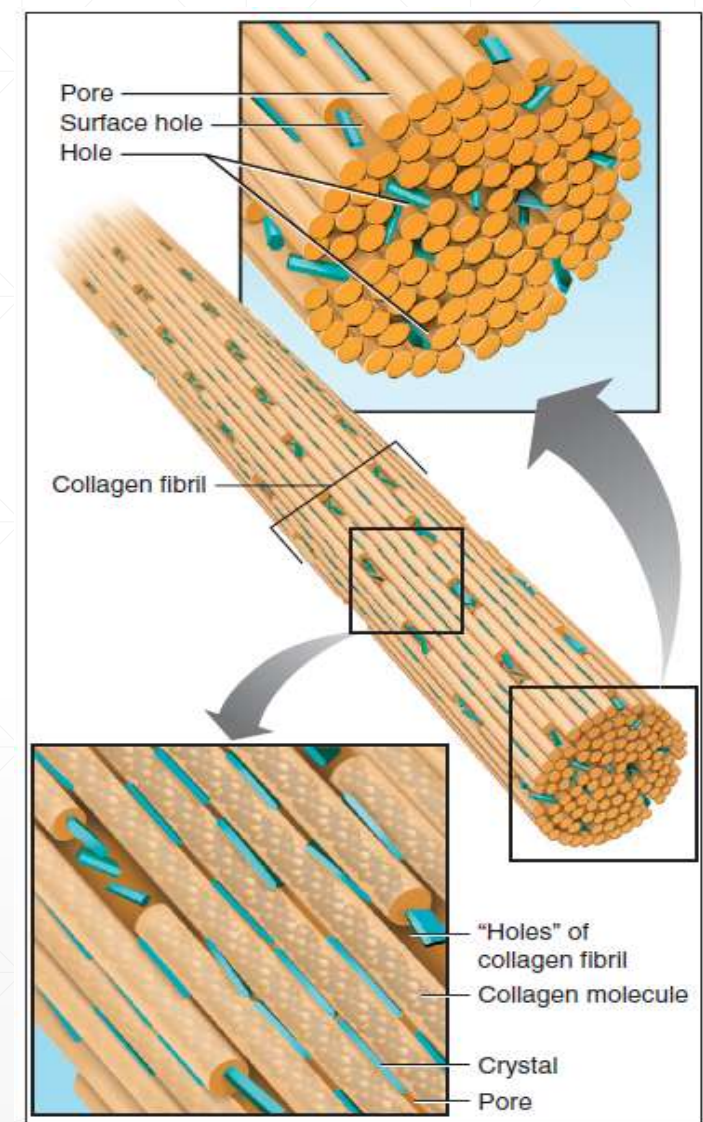
Localization of DPP at mineralization front

MINERALIZATION BY HETEROGENOUS NUCLEATION

- The crystals are formed in the collagen, the noncollagen proteins regulate crystal formation along with regulating crystal type, formation rate, size and shape.
 - Noncollagenous proteins can bind selectively to different surfaces of the crystal, preventing further growth and thereby determining the final size of the crystal.
 - As matrix deposition and mineralisation continue, there will always be a zone of mineralisation discernible histologically.
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MINERALIZATION BY HETEROGENEOUS NUCLEATION

- Mineral crystals first appear in the hole zone formed by the quarter-stacking of the collagen molecules. The collagen triple helices are stacked in parallel with overlapping regions and spaces between adjacent molecules.
- Ultrastructurally, the bulk of the dentine mineral is on the surface of the collagen.



Secretion of collagen and non-collagenous proteins



Formation of Predentine



Collagen fibrils extend to the mineralization front



Secretion of DPP near the mineralization front



DPP binds to pre-existing collagen fibrils



DPP attracts Ca²⁺ ions



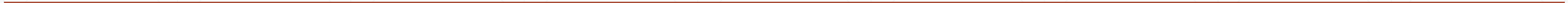
Nucleation of hydroxyapatite crystals



Regulation of crystal growth along collagen fibrils



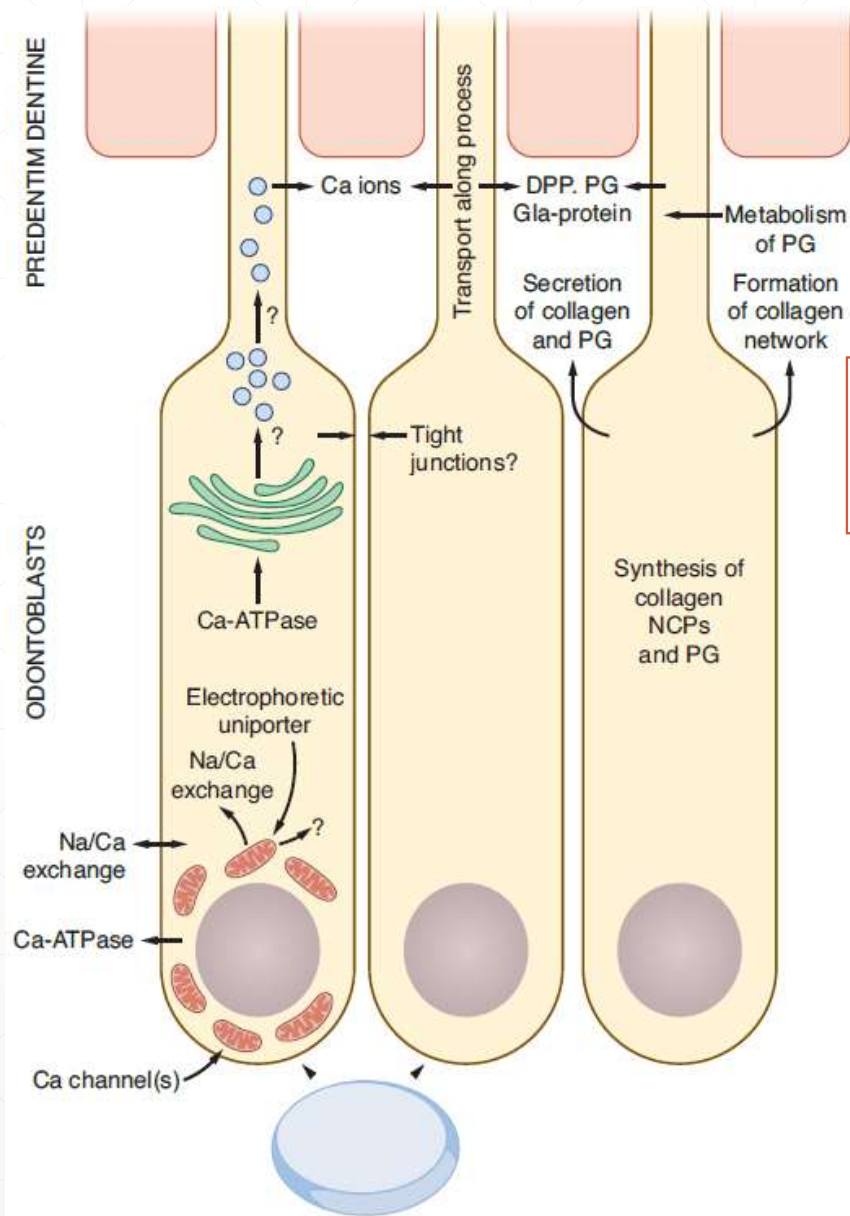
Mineralization of dentine



Na/Ca exchanger helps regulate intracellular calcium.

Ca-ATPase pumps calcium using ATP

Ca channels allow calcium to enter the odontoblast.



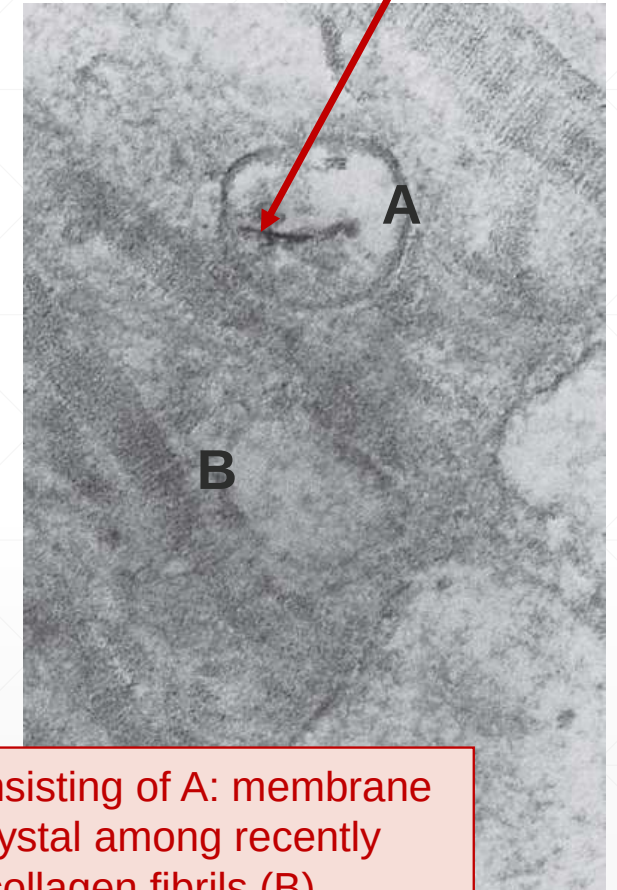
DPP and proteoglycans are transported along the odontoblast process

Odontoblast cell body secretes the collagen fibrils to form meshwork

Odontoblast cell body synthesizes type I collagen, non-collagenous proteins, proteoglycans

MINERALIZATION BY MATRIX VESICLES

- The matrix vesicle is a small, membrane-bound structure that buds off from the cell to form an independent unit within the first-formed organic matrix of hard tissues.
- Mineral crystals develop within the vesicles, they rupture and release initial mineral spheres (calcospherites) and they have a role in initial mineralization.



Matrix vesicle consisting of A: membrane containing a crystal among recently deposited collagen fibrils (B)

Cell budding



Matrix vesicle formation



Accumulation of Ca^{2+} and PO_4^{3-} ions



High local concentration of ions



Formation of initial hydroxyapatite crystals



Crystal growth within a vesicle



Vesicle membrane ruptures



Crystals spread into the surrounding collagen matrix



Calcospherite formation and mineralization

Matrix vesicles initiate the earliest mineralization in mantle dentine, while heterogeneous nucleation on collagen fibrils is the principal mechanism for subsequent and widespread dentine mineralization, especially in circumpulpal dentine.

Formation of Calcospherites

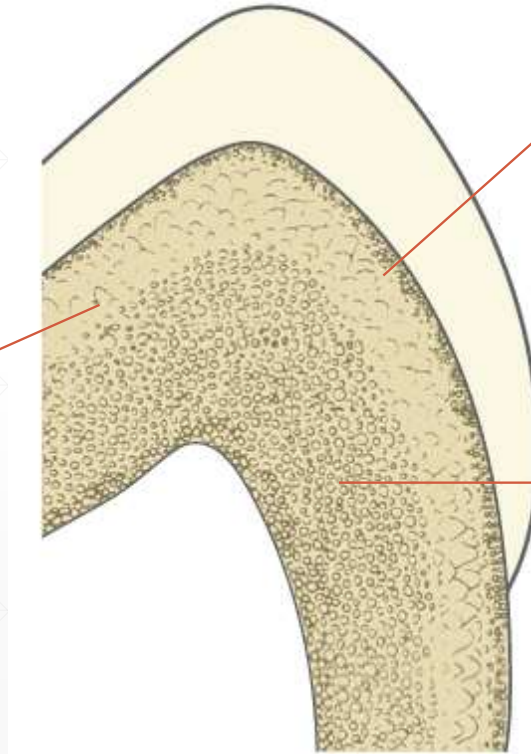
- Calcospherites are spherical mineralization centers formed by the growth and aggregation of hydroxyapatite crystals. They enlarge and fuse to produce mineralized dentine matrix.
- The pattern of calcospherite formation is common throughout dentine.
- The size and shape seems to be governed by the rate of dentine formation. They may be spherical or arcade shaped.



Calcospherites (arrowed) at the mineralising front of dentine

Formation of Calcospherites

In the middle region of the circumpulpal dentine they are larger, more widely spaced and arcade.

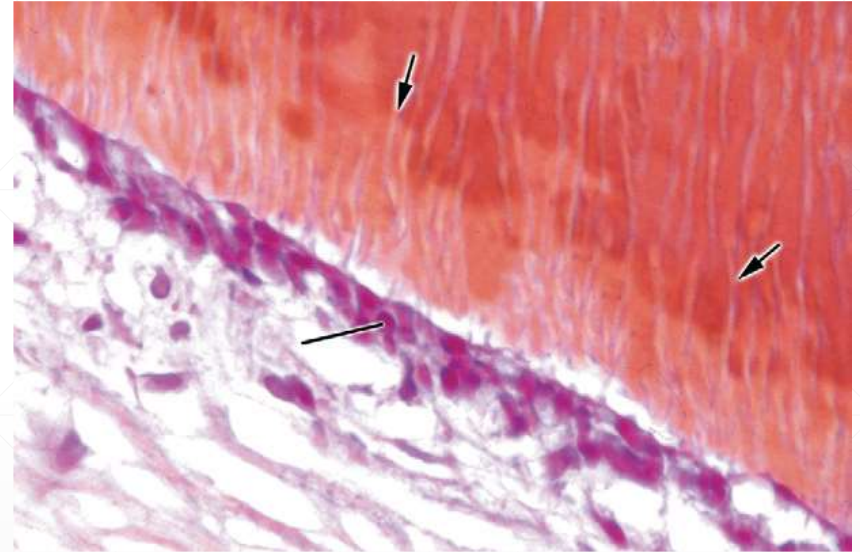


In the mantle dentine and superficial circumpulpal dentine they are small, spherical and closely packed.

The inner half to two-thirds of the circumpulpal dentine contains spherical ones.

GLOBULAR PATTERN OF MINERALIZATION

- Globular calcification involves the deposition of crystals and with continued crystal growth, globular masses are formed that continue to enlarge and eventually fuse to form a single calcified mass.
- This pattern of mineralization is best seen in the mantle dentin region, where matrix vesicles give rise to mineralization foci that grow and coalesce. The largest globules deposit where dentine deposition is fastest.



Low-magnification photomicrographs of the predentin-dentin interface illustrating globular pattern.

LINEAR PATTERN OF MINERALIZATION

- When the rate of formation progresses slowly, the mineralization front appears more uniform and the process is said to be linear.



Light photomicrographs of the predentin-dentin interface illustrating linear pattern.

Mineralization of Mantle Dentine

Initiated by matrix vesicles released from the cell budding of odontoblast processes. The hydroxyapatite crystals first form within these vesicles, enlarge, and spread into the surrounding collagen matrix. The mineralization front is typically globular, resulting in the formation and fusion of calcospherites.

Mineralization of Circumpulpal Dentine

Mainly by heterogeneous nucleation, hydroxyapatite crystals are deposited on collagen fibrils, within the hole (gap) zones.

DPP promotes crystal nucleation and growth at the mineralization front. Mineralization pattern may be globular or linear, and failure of calcospherite fusion results in interglobular dentine (outer part of circumpulpal).

CLINICAL CONSIDERATION

Dentinogenesis imperfecta (types I, II, and III) are inherited disorders affecting dentine formation. Types II and III are caused by mutations in the DSPP gene. Dentine formed is abnormal and is deposited excessively within the pulp chambers and poorly attached to the enamel.

-Clinical appearance: Malformed teeth, dark in color, bulbous crowns, loss of translucency. The enamel appears normal but flakes off.

-Radiographic appearance: cervical constriction and obliteration of the pulp chambers and canals.



THANK YOU
